Yiqiang Zhou

Email: cloudyu1215@outlook.com Github: https://github.com/Cloudyu1215 Google Scholar

Research Interests: Low-level Vision, Image Enhancement & Restoration.

EDUCATION

Northeast Forestry University Harbin, China, 09/2024 — Present
Master of Engineering in Computer Science Cumulative GPA: 88.96/100

Northeast Forestry University Harbin, China, 09/2019 — 06/2023
Bachelor of Engineering in Computer Science

RESEARCH EXPERIENCE

Institute of Artificial Intelligence (TeleAI), China Telecom Shanghai, China
*Full-time Researcher, Water-related Embodied AI Group — **Homepage***
*Under the guidance of **Prof. Xuelong Li** (Member of Academia Europaea; IEEE/ACM Fellow) — **Homepage*** 10/2025 — Present

- Designed an ultra-lightweight underwater image enhancement framework that balances restoration quality and inference efficiency, and deployed it for real-time execution on edge devices; validated the system in field experiments at Qiandao Lake, with outcomes reported in Github and submitted to *IEEE Transactions on Image Processing (TIP)*.
- Planned and developed a simulated underwater sediment degradation dataset, and designed a corresponding image restoration pipeline for turbid underwater environments.
- Co-developed a large-scale underwater robotic simulation strategy from sim-to-sim to sim-to-real, including reward function design and transfer to real-world deployment.
- Developed ISP pipelines for industrial cameras and telephoto lenses, and deployed a real-time single-image dehazing algorithm for daytime scenarios.
- Contributed to the development of a real-time underwater fish detection algorithm for aquatic perception and monitoring tasks.

PUBLICATIONS

[1] **Yiqiang Zhou**, Yifan Chen, Zhe Sun, Jijun Lu, Ye Zheng, and Xuelong Li*. "Advancing Visual Reliability: Color-Accurate Underwater Image Enhancement for Real-Time Underwater Missions." arXiv preprint arXiv:2603.16363, 2026. Link: https://arxiv.org/abs/2603.16363. Code: https://github.com/Cloudyu1215/UIE. **(Under review at *IEEE Transactions on Image Processing (TIP), IF=13.7*)**

[2] **Yiqiang Zhou**, Xindan Gao*, Jifeng Guo, Guang Li, Lu Wang, and Jing Liu. "LUXFormer: Low-Light Image Enhancement via Joint Spatial-Frequency Illumination Modeling." *Journal of King Saud University Computer and Information Sciences*, vol. 37, no. 8, p. 251, 2025. doi: https://doi.org/10.1007/s44443-025-00270-5. **(Published, IF=6.1)**

[3] **Yiqiang Zhou**, Xindan Gao*, Jifeng Guo, Yanfeng Qiao, Yan Chen, Guang Li, and Lu Wang. "ClearNight: Synergistic Dual-Prior Aggregation for Nighttime Image Dehazing." **(Under review at *Pattern Recognition*, IF=8.6)**

[4] Jijun Lu, **Yiqiang Zhou**, et al. (corresponding author: Xuelong Li). "Compressive sensing-inspired self-supervised single-pixel imaging." arXiv preprint arXiv:2603.29732, 2026. Link: https://arxiv.org/abs/2603.29732. **(Under review at *Advanced Imaging*)**

[5] Guang Li , **Yiqiang Zhou**, and Xindan Gao*. "RGB-T Fusion Network for Low-Light Scene Segmentation." *Computer Engineering*, pp. 1–14, 2026. **(Published)**

AWARDS

- **National Scholarship for Graduate Students (Top 0.1%, National Level)**
  Awarded by the Ministry of Education for exceptional academic and research performance Oct 2025
- **Outstanding Volunteer, China Western Program (National Level)**
  Recognized for contributions to rural education and public service Oct 2024
- **Outstanding Individual in Rural Revitalization Program (Provincial Level)**
  Recognized as a core contributor to rural development initiatives Dec 2024
- **First-class Academic Scholarship (Top 1%, University Level)**
  Awarded for outstanding academic performance Oct 2025
- **Merit Student (University Level)**
  Awarded for academic excellence and overall performance Oct 2025
- **Outstanding Student Leader (University Level)**
  Recognized for leadership and service in student organizations Jun 2025

VOLUNTEER & SERVICE

China Western Program Gansu, China
Volunteer Teacher Jul 2023 – Aug 2024

- Served as a volunteer teacher in rural Gansu through the China Western Program, teaching computer science, physical education, and biochemistry, while developing course materials tailored to students with diverse academic backgrounds and limited resources.
- Provided classroom instruction, after-class mentoring, and extracurricular support to enhance students' academic engagement and overall development, contributing to improved access to education in underserved communities.

SKILLS

Technical Skills: Python, MATLAB, PyTorch, Linux, TensorRT & ONNX, Git & Docker
Languages: Mandarin (native), English (IELTS scheduled, experienced in academic writing)